Calibrated Confidence for Kilometre-Scale Feed-Forward SLAM

Mid-Submission Report, Mobile Robotics (Monsoon 2026)

Kavish Vora Pranav Srivastava Mehrish Khan (IIIT Hyderabad)

1 Problem statement

Feed-forward 3D models such as VGGT [1] take a few dozen images and, in one forward pass, return camera poses, depth and point maps, and a per-pixel confidence $C \geq 1$. GPU memory limits one pass to about 30 to 100 images, so VGGT-Long [2] runs VGGT on overlapping windows of a long monocular drive and joins them in a pose graph. Every window k is a node with an unknown placement $S_k \in \mathrm{Sim}(3)$ (rotation, translation, scale). Every edge (i, j) is a measured relative placement Z_{ij}: a *seam* between neighbouring windows, from the Sim(3) that aligns the 3D points of their shared frames, or a *loop* between two visits of the same place. The trajectory is the solution of

$$\{S_k^\star\} = \arg\min_{\{S_k\}} \sum_{(i,j)} \rho\Big(r_{ij}^\top \Lambda_{ij}\, r_{ij}\Big), \qquad r_{ij} = \mathrm{Log}\Big(Z_{ij}^{-1} S_i^{-1} S_j\Big) \in \mathbb{R}^7, \tag{1}$$

where ρ is an optional robust kernel and Λ_{ij} is the 7×7 information matrix of the edge. VGGT-Long sets $\Lambda_{ij} = I$ for every edge, although some edges are clearly worse than others (low texture, turns, little overlap) and VGGT already reports how confident it is in every pixel.

Our problem. Learn, without any training of the network, a map g from confidence statistics ϕ_{ij} of an edge to a diagonal information matrix $\Lambda_{ij} = \mathrm{diag}(\sigma_{\mathrm{rot}}^{-2} I_3, \sigma_{\mathrm{trans}}^{-2} I_3, \sigma_{\mathrm{scale}}^{-2})$ that is *calibrated*, i.e. $\mathbb{E}[e_{ij,b}^2 \mid \phi_{ij}] \approx \sigma_b^2(\phi_{ij})$ for each block b, where e_{ij} is the true 7-number error of the edge. We ask three questions: (Q1) does pixel confidence predict depth error, and can it be calibrated; (Q2) does it predict the error of seams and loops; (Q3) do calibrated edge weights give a more accurate trajectory than equal weights? Everything that is fitted is fitted on KITTI [21] sequences 05 and 07; sequences 00 and 02 are held out and only used for scoring.

Refinements since the proposal. (i) VGGT-Long and our schemes measure the edge residual differently (the order of the product inside Log), so we add an equal-weight control in our convention, (a′), next to VGGT-Long's own (a). (ii) True edge errors e_{ij} come from ground-truth placements of each window; we found that KITTI's ground truth is not consistent in height at the loop revisits of 02 (and less so 00), so loop labels there measure ground-truth error and are reported separately. (iii) We use windows of 32 frames with 16 frames overlap, chosen by a GPU memory sweep, close to the paper's smallest setting.

2 Literature review

Feed-forward reconstruction. DUSt3R [4] and MASt3R [5] regress dense point maps with confidence for image pairs; VGGT [1] handles many views at once and also predicts cameras. CUT3R [10] and StreamVGGT [11] process video as a stream with a persistent memory, and VGGT-Ω [12] adds scene-summarising register tokens. These models are trained to output confidence with a loss of the form $Ce - \alpha \log C$ [6], which is minimised at $C = \alpha/\mathbb{E}[e]$; a perfectly trained model would therefore have error proportional to $1/C$. This gives a concrete, testable expectation for Q1.

Long sequences with feed-forward models. VGGT-Long [2] is our baseline: chunking, confidence-weighted Sim(3) alignment, SALAD [20] loop retrieval on DINOv2 [19] features, and a Sim(3) pose graph with equal weights. VGGT-SLAM [13] aligns submaps with 15-DoF SL(4) transforms for uncalibrated cameras; our weighting applies there unchanged. MASt3R-SLAM [14] uses MASt3R's confidence as weights in its back end without checking whether it is calibrated, and DROID-SLAM [15] learns its weights end to end, which needs training. Sim(3) pose graphs for monocular

scale drift go back to Strasdat et al. [3], and ORB-SLAM3 [25] is the standard classical monocular baseline.

Is confidence trustworthy? Hillemann et al. [7] study the uncertainty quality of VGGT on DTU. The calibration audit of N. Deng et al. [8] checks native confidence of seven feed-forward models on 13 datasets including KITTI, finds that confidence ranks errors well but understates their size, and proposes a two-constant power-law correction; we use it as baseline (g). Both work per pixel. GeoCond [9] (WACV 2027) is the closest work to ours: it observes that pose failures of feed-forward models track geometric conditioning (overlap, parallax, rotation) that native confidence captures poorly, trains a small adapter that outputs pose-level uncertainty, and uses it for gated refinement and pose-graph weighting. Our project differs in scope: we do not train anything, we target the Sim(3) window-to-window seams and loop closures of a kilometre-scale pipeline including scale, and we measure trajectory error over whole drives. GeoCond's finding also motivates our geometry-only baseline (d).

Calibration and robust back ends. Kuleshov et al. [16] measure the calibration of regression models with reliability diagrams and fix it with isotonic regression, which we follow. Dynamic Covariance Scaling (DCS) by Agarwal et al. [17] protects pose graphs against wrong loop closures; we use it and the Huber kernel on loop edges. Trajectories are scored with Sim(3)-aligned ATE [18, 23], checked against the evo tool [24].

Why the project is viable and worth doing. The audits above show that pixel confidence ranks errors but is miscalibrated; GeoCond shows that pose-level reliability can improve graph weighting when it is learned. Nobody has tested whether VGGT's native confidence, calibrated without training, is a reliable weight for the edges of a long Sim(3) pose graph. All parts are available and cheap: VGGT and VGGT-Long are open, KITTI provides ground-truth poses and LiDAR, and the calibration is fitted on a CPU in seconds.

3 Progress so far

Pipeline and runs (all on one RTX A4000, 16 GB). (1) KITTI 00, 02, 05, 07: images, ground truth converted from camera 0 to the colour camera 2 that VGGT sees, and all 13,064 LiDAR scans projected onto VGGT's 518×154 input grid. (2) VGGT-1B in bfloat16 on 814 windows plus 43 loop windows; a memory sweep (7.7 to 12.2 GB for 8 to 64 frames, about 17 frames/s) and four automatic convention checks before the long runs. (3) A re-implementation of VGGT-Long's back end in which edge weights can be changed, checked against VGGT-Long's code (same alignment to 10^{-15}, same loop pairs, same optimum), and the original `vggt_long.py` run at four window sizes on all four drives. (4) The true error of every seam and loop from ground-truth placements of each window (834 of 853 edges usable; windows where the car is nearly stopped are skipped). (5) A pixel audit of both VGGT heads against LiDAR (478 million confidence/error pairs). (6) Calibration on 05/07 and eight weighting schemes on all four drives, each without loops, with loops, and with Huber or DCS on loop edges. (7) Checks: 44 automated tests, ATE equal to evo on all 72 trajectories (largest difference 10^{-9} m), and a LiDAR check of the ground truth at every loop revisit.

Weighting schemes. (a) VGGT-Long's equal weights; (a′) equal weights in our residual convention, the control for (b) to (g); (b) one fitted variance each for rotation, translation (relative to the window's depth) and scale, with a separate factor for loops; (c) (b) scaled by the squared mean confidence of the edge; (d) isotonic map from the alignment residual (geometry only); (e) **ours**: isotonic map from the edge's mean point confidence; (f) an oracle using each edge's measured error (a reference, not a method); (g) the per-pixel power law of [8], summed over the edge's pixels. Each predicted variance σ_b^2 becomes the weight $1/\sigma_b^2$ of its block.

3.1 Reproduction of VGGT-Long

The original code reproduces the paper within 0.5% at 30/15 on all four drives, and our re-implementation matches the original code within 0.7% at 32/16 (Figure 1), so our baseline is VGGT-Long itself. The very large error on 02 with small windows is also in the paper. At 60/30

Table 1: ATE (m). Paper: Table 1 of [2]. Original: `vggt_long.py` run by us. Ours (a): our re-implementation with equal weights. "–": not run.

Source	W/O	05	07	00	02	Source	W/O	05	07	00	02
Paper	30/15	10.20	4.00	9.06	99.95	Paper	60/30	9.15	2.68	8.06	34.16
Original	30/15	10.24	3.98	9.07	99.65	Original	60/30	9.85	2.66	8.63	52.78
Original	32/16	10.04	4.09	9.13	104.61	Paper	90/45	9.88	3.47	11.97	49.85
Ours (a)	32/16	**10.02**	**4.06**	**9.18**	**105.07**	Original	90/45	–	–	–	50.56

three drives are within 8% of the paper. On 02 at 60/30 we get 52.78 m against the paper's 34.16 m, although the same code matches the paper on 02 at 30/15 and 90/45; an October 2025 version of the code, 200 instead of 30 optimiser iterations and the older loop threshold all give exactly 52.78 m, so we could not reproduce this one number.

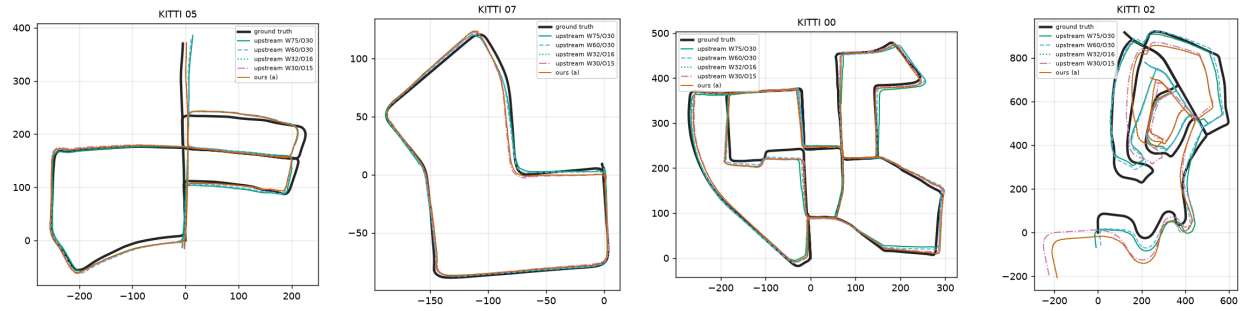


Figure 1: Top view of the trajectories (m) after Sim(3) alignment: ground truth (black), the original VGGT-Long code at four settings, and our re-implementation (a) at 32/16, which lies on top of the original code at the same setting.

3.2 Q1: pixel confidence against LiDAR

Table 2: Held-out sequences (00, 02). Slope of log(error) against log(confidence) (-1 would match error $\propto 1/C$). Calibration error: 0 is perfect. Calibration fitted on 140 million pixels of 05/07.

Head	Pixels	Slope	Spearman	Raw κ/C	Power law [8]	Isotonic
Depth head	337.9 M	-0.70	-0.34	0.40	0.25	**0.18**
Point head	337.9 M	-0.63	-0.40	0.60	0.53	**0.32**

Confidence ranks pixel errors correctly but the error falls more slowly than $1/C$ (slope about -0.7). A monotone calibration learned on other sequences removes most of the mismatch (0.40 to 0.18 for the depth head; Figure 2). The largest errors sit on object borders, which also get low confidence (Figure 5, appendix).

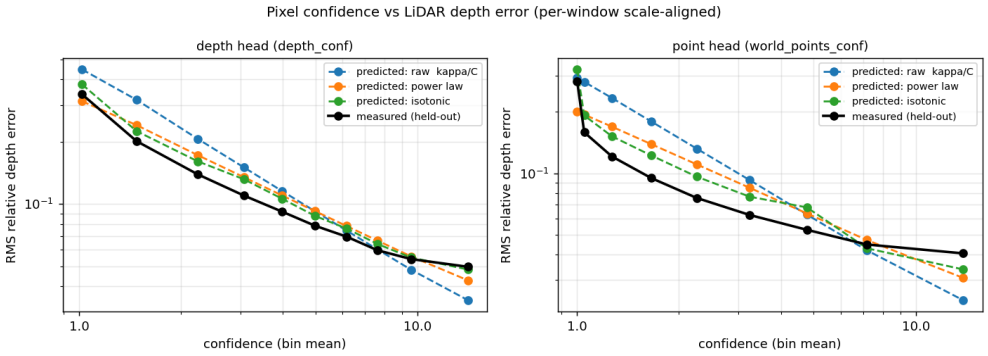


Figure 2: Root-mean-square depth error per confidence bin on 00 and 02 (black) and what each predictor expects.

3.3 Q2: seams and loops

Table 3: Predicted against true error on held-out edges. Ratio: mean squared error over mean predicted variance (1 is perfect). NEES: 7 is perfect. Coverage: share of edges inside the predicted 95% region (0.95 is perfect). Right: Spearman correlation of a seam feature with the seam's true error (569 held-out seams).

Edges	Scheme	rot	trans	scale	NEES	Cov.	Feature	rot	trans
569 seams	(b) uniform	1.36	1.32	1.29	9.3	0.80	Mean confidence	−0.44	−0.12
	(c) raw conf.	0.90	0.87	0.85	7.7	0.86	Median confidence	−0.26	−0.13
	(d) geometry	1.44	1.40	1.33	11.0	0.77	Alignment residual	0.24	**0.55**
	(e) calibrated	1.29	1.26	1.29	11.5	0.76	Inlier fraction	−0.17	−0.25
	(g) power law	1.57	1.07	1.04	9.9	0.78			
27 loops	(b) uniform	2.00	22.36	9.78	82.9	0.11			
	(e) calibrated	1.58	23.27	10.70	119.8	0.15			

Seams. All schemes predict the size of seam errors within a factor of about 1.5. A seam's mean confidence ranks its rotation error (Spearman −0.44) but hardly its translation error (−0.12); the alignment residual ranks translation best (0.55), in line with GeoCond's observation that geometry matters for pose reliability. The worst seams see vegetation or are taken during turns, where confidence is low almost everywhere (Figure 6).

Loops. All 43 loops are real revisits, yet on held-out loops the squared errors are about 20 times the prediction (about 4.7 times in error size). The cause is mostly the ground truth (Table 4): on 02 the ground truth places the two visits of the same road 1 to 4 m apart *in height*. Placing both visits' LiDAR scans with the ground-truth poses puts their road surfaces 0.8 to 4.1 m apart at the same spot on 02, up to 1.3 m on 00, and within 0.2 m on 07 (0.5 m for two loops of 05). A road cannot have two heights, so the ground-truth heights are inconsistent at these revisits and VGGT is right to place the visits together (Figure 7). Seam labels are not affected, because each seam only uses the ground truth within one short window.

Table 4: Loops per sequence (medians, maxima in brackets). Ground-truth (GT) gap between the two visits, its vertical part, and the height difference between the two visits' LiDAR road surfaces after GT placement (0 if GT is consistent).

Seq.	Loops (usable)	GT gap (m)	GT vertical (m)	LiDAR road gap (m)	Edge error (m)
05	12 (10)	0.31 (0.81)	0.05 (0.50)	0.05 (0.52)	0.37 (1.13)
07	3 (1)	0.41 (0.52)	0.12 (0.16)	0.18 (0.21)	0.97 (0.97)
00	20 (19)	0.67 (2.06)	0.47 (1.07)	0.49 (1.30)	1.16 (2.15)
02	8 (8)	2.64 (4.09)	1.94 (4.01)	2.02 (4.06)	3.05 (6.18)

3.4 Q3: trajectories

Table 5: ATE (m) / KITTI translation drift t_rel (%), loops on, window 32/16. *Uses the measured edge errors.

Weighting	05 (fit)	07 (fit)	00 (test)	02 (test)
No loops (any scheme)	48.70 / 13.78	22.36 / 13.30	140.49 / 34.47	181.52 / 40.90
(a) VGGT-Long	10.02 / 5.11	4.06 / 3.94	9.18 / 4.82	**105.07 / 26.95**
(a′) equal, control	9.80 / 5.05	4.07 / 3.93	9.14 / 4.74	106.90 / 27.45
(b) uniform	11.52 / 5.83	**3.61 / 3.53**	10.15 / 4.88	113.36 / 29.81
(c) raw confidence	10.05 / 5.12	4.55 / 3.84	10.28 / 5.03	119.60 / 31.44
(d) geometry only	11.10 / 5.49	4.49 / 3.88	9.34 / 4.75	113.27 / 29.86
(e) calibrated (ours)	9.57 / 4.78	4.37 / 3.83	**9.09 / 4.68**	117.07 / 30.59
(f) oracle*	**8.94 / 4.30**	4.59 / 4.17	11.92 / 5.31	116.08 / 29.56
(g) power law	10.39 / 5.24	4.37 / 3.79	9.89 / 4.89	119.70 / 31.47

Closing loops cuts the error 5 to 15 times (00: 140 m to 9 m). Edge weights matter much less: against

the fair control (a′), our calibrated scheme (e) is better on 05 (9.57 against 9.80 m) and 00 (9.09 against 9.14 m) but worse on 07 and 02; on a given sequence the schemes differ by 10 to 25%, and the residual convention alone ((a) against (a′)) changes ATE by up to 2%. On 02 every weighted scheme is worse than equal weights; we have not yet isolated why. Huber never down-weighted a loop; DCS switched off all 8 true loops on 02 for schemes (b), (c), (d), (g) (ATE about 175 m). The oracle is not the best either, which only shows that inverse measured-error weighting is not an upper bound here.

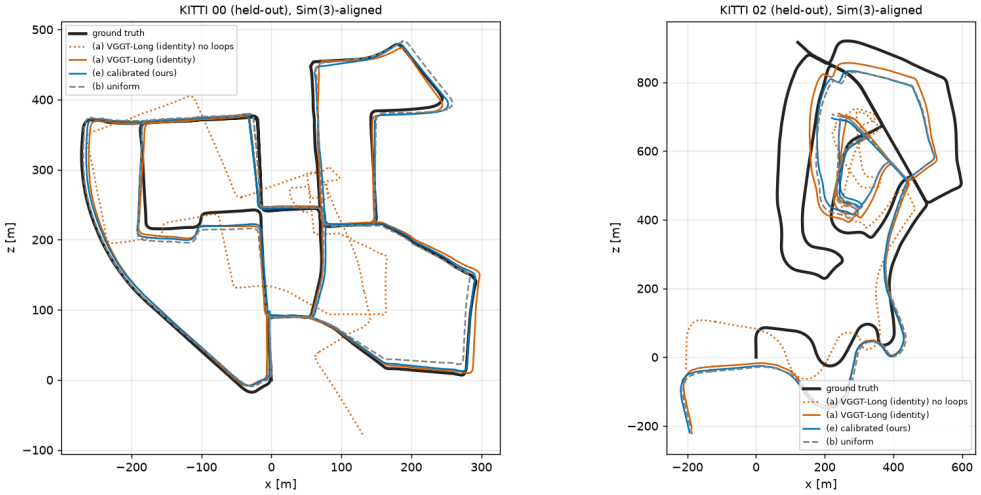


Figure 3: Held-out trajectories (top view, m). On 00 loops pull the drifting chain (dotted) onto the ground truth. On 02 all loops join the start of the drive to its end, so the long middle section keeps its drift.

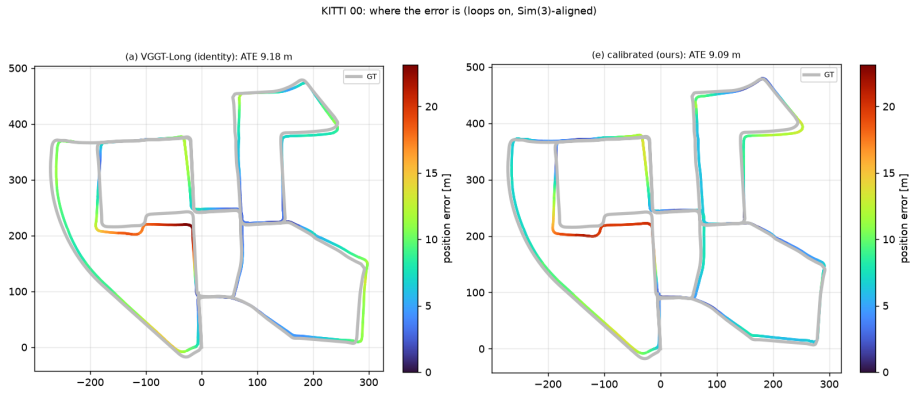


Figure 4: KITTI 00, position error along the path with VGGT-Long's weights (left) and ours (right). The largest error (about 20 m) is on a detour that no loop closes.

Summary. VGGT-Long is reproduced; pixel confidence ranks depth error and is well predicted after calibration (Q1); averaging it over a seam keeps information about rotation but loses most of it about translation (Q2); KITTI's ground truth is inconsistent in height at the 02 revisits, which affects loop labels; and calibrated edge weights so far give small gains on some sequences and losses on others (Q3).

4 Timeline for the rest of the project

Table 6: Remaining work, as planned in the proposal. Final submission Nov 23, demo Nov 26.

Dates	Work
Nov 6 to Nov 9	False-loop stress test: add 0, 1, 2, 4, 8, 16 wrong loops between distant windows (20 random seeds) and measure which schemes and robust kernels survive.
Nov 10 to Nov 14	Waymo Open Dataset [22] and TUM RGB-D [23] loaders and runs; ORB-SLAM3 (monocular) [25] as a classical reference.
Nov 15 to Nov 19	Bonus: streaming models CUT3R [10] and StreamVGGT [11], and VGGT-Ω [12], in the same pipeline; memory and speed table.
Nov 20 to Nov 22	MASt3R-SLAM [14] if time allows; final figures and report; code freeze Nov 22.
Nov 23 / Nov 26	Final submission / demo.

References

[1] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, "VGGT: Visual geometry grounded transformer," in *Proc. CVPR*, 2025.

[2] K. Deng, Z. Ti, J. Xu, J. Yang, and J. Xie, "VGGT-Long: Chunk it, loop it, align it: Pushing VGGT's limits on kilometer-scale long RGB sequences," arXiv:2507.16443, 2025.

[3] H. Strasdat, J. M. M. Montiel, and A. J. Davison, "Scale drift-aware large scale monocular SLAM," in *Proc. RSS*, 2010.

[4] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, "DUSt3R: Geometric 3D vision made easy," in *Proc. CVPR*, 2024.

[5] V. Leroy, Y. Cabon, and J. Revaud, "Grounding image matching in 3D with MASt3R," in *Proc. ECCV*, 2024.

[6] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?," in *Proc. NeurIPS*, 2017.

[7] M. Hillemann, R. Langendörfer, S. Landgraf, and M. Ulrich, "Uncertainty quality of VGGT: An analysis on the DTU benchmark dataset," *ISPRS Annals*, vol. XI-2-2026, pp. 665–672, 2026.

[8] N. N. Deng, Q. Cheng, N. Zeller, and D. Cremers, "A calibration audit of confidence in feed-forward 3D reconstruction models," arXiv:2608.29705v3, 2026.

[9] D. Ahmedt-Aristizabal, M. A. Armin, R. Tsuchida, and L. Petersson, "GeoCond: A conditioning-aware reliability adapter for feed-forward 3D reconstruction," in *Proc. WACV*, 2027. arXiv:2609.18465.

[10] Q. Wang, Y. Zhang, A. Holynski, A. A. Efros, and A. Kanazawa, "Continuous 3D perception model with persistent state," in *Proc. CVPR*, 2025.

[11] D. Zhuo, W. Zheng, J. Guo, Y. Wu, J. Zhou, and J. Lu, "Streaming 4D visual geometry transformer," arXiv:2507.11539, 2025.

[12] J. Wang, M. Chen, S. Zhang, N. Karaev, J. Schönberger, P. Labatut, P. Bojanowski, D. Novotny, A. Vedaldi, and C. Rupprecht, "VGGT-Ω," in *Proc. CVPR*, 2026. arXiv:2605.15195.

[13] D. Maggio, H. Lim, and L. Carlone, "VGGT-SLAM: Dense RGB SLAM optimized on the SL(4) manifold," in *Proc. NeurIPS*, 2025.

[14] R. Murai, E. Dexheimer, and A. J. Davison, "MASt3R-SLAM: Real-time dense SLAM with 3D reconstruction priors," in *Proc. CVPR*, 2025.

[15] Z. Teed and J. Deng, "DROID-SLAM: Deep visual SLAM for monocular, stereo, and RGB-D cameras," in *Proc. NeurIPS*, 2021.

[16] V. Kuleshov, N. Fenner, and S. Ermon, "Accurate uncertainties for deep learning using calibrated regression," in *Proc. ICML*, 2018.

[17] P. Agarwal, G. D. Tipaldi, L. Spinello, C. Stachniss, and W. Burgard, "Robust map optimization using dynamic covariance scaling," in *Proc. ICRA*, 2013.

[18] S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE TPAMI*, 13(4):376–380, 1991.

[19] M. Oquab, T. Darcet, T. Moutakanni, *et al.*, "DINOv2: Learning robust visual features without supervision," *TMLR*, 2024.

[20] S. Izquierdo and J. Civera, "Optimal transport aggregation for visual place recognition," in *Proc. CVPR*, 2024.

[21] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proc. CVPR*, 2012.

[22] P. Sun, H. Kretzschmar, X. Dotiwalla, *et al.*, "Scalability in perception for autonomous driving: Waymo open dataset," in *Proc. CVPR*, 2020.

[23] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of RGB-D SLAM systems," in *Proc. IROS*, 2012.

[24] M. Grupp, "evo: Python package for the evaluation of odometry and SLAM," `https://github.com/MichaelGrupp/evo`, 2017.

[25] C. Campos, R. Elvira, J. J. Gómez Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM," *IEEE Trans. Robotics*, 37(6):1874–1890, 2021.

A Additional figures

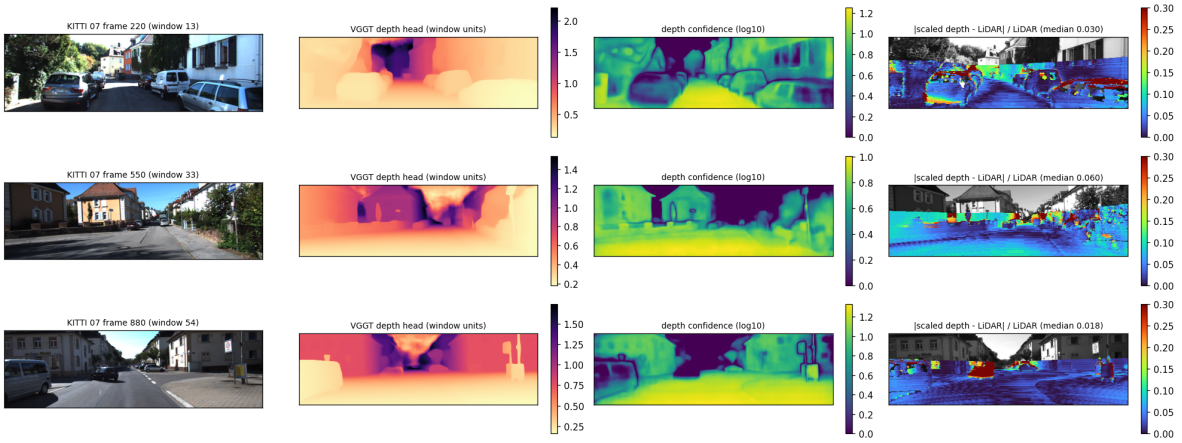


Figure 5: KITTI 07: image, VGGT depth, depth confidence (log), and relative depth error at LiDAR pixels after scale alignment (median 2 to 6%, high on object borders).

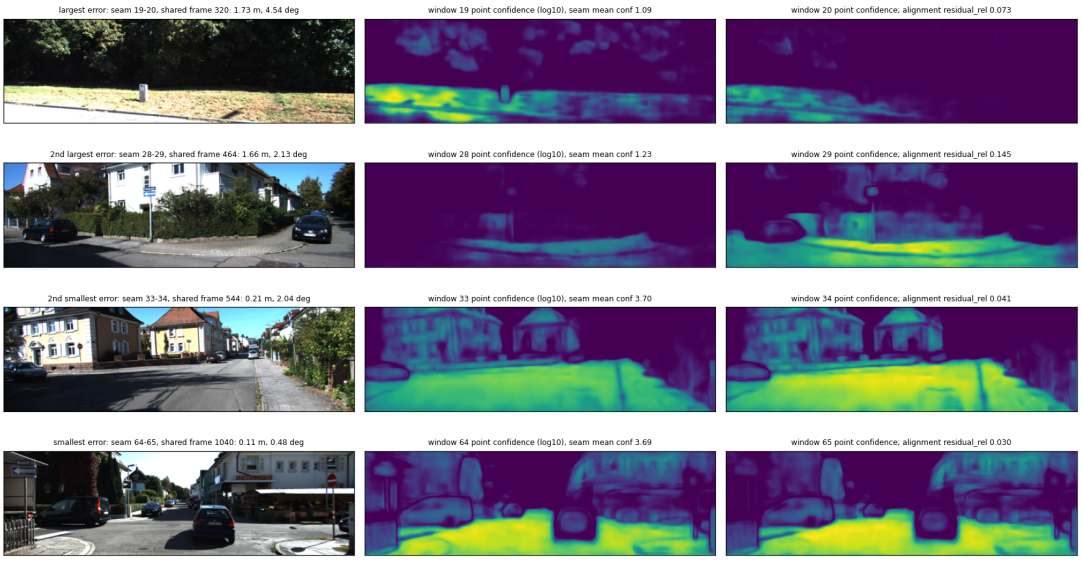


Figure 6: KITTI 07, the two worst and two best seams by true error: shared frame and point confidence of the two windows.



Figure 7: KITTI 02, loops with the largest and smallest measured error. Top: the ground truth places the two visits 3.09 m apart, almost all in height; their LiDAR road surfaces, placed with the ground truth, are 3.09 m apart in height.